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Niggli Reduction Reconsidered

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Abstract

Niggli reduction of a unit cell is conventionally described as a sequence of conditional steps enforcing inequalities on the components of the metric tensor in $\mathbf{G}^6$. We argue that the essential geometric content of Niggli reduction is the minimization of $\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3 = a^2 + b^2 + c^2$, the sum of squared edge lengths, followed by secondary conditions that select a unique canonical representative from among all cells sharing the minimum trace. This reframing clarifies the termination argument of Křivý & Gruber (1976), whose case analysis is shown to be implicitly a trace-minimization argument. The distinct causes of infinite loops in floating-point implementations are enumerated, and the treatment of Grosse-Kunstleve *et al.* (2004) is examined in this light. Multiple distinct cells can share the same minimum trace value, as illustrated by example; Niggli's secondary conditions are essential precisely because of this degeneracy.

1. Introduction

Niggli reduction (Niggli, 1928) produces a canonical primitive unit cell. The computational algorithm of Křivý & Gruber (1976) is widely used for this purpose.

The conventional description of Niggli reduction is procedural: a sequence of steps that enforce inequalities on the components of the metric tensor in $\mathbf{G}^6$. Here we describe Niggli reduction as minimization of a single quantity, $\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3$, followed by secondary conditions that select a unique cell from among those sharing the minimum trace. To our knowledge this description does not appear in the literature. Section 2 gives the historical background. Section 3 defines $\mathbf{G}^6$. Section 4 develops the trace-minimization description. Section 5 enumerates the causes of infinite loops in floating-point implementations and examines Grosse-Kunstleve *et al.* (2004). Section 6 gives examples. Section 7 is the summary.

2. Historical Background

Niggli reduction has its mathematical roots in the theory of positive definite ternary quadratic forms. As noted by Roof (1967), the conditions defining a reduced cell descend from the work of Eisenstein (Eisenstein, 1851), who established criteria for a canonical reduced form of a positive definite ternary quadratic — precisely the object represented by the metric tensor in $\mathbf{G}^6$. Niggli (Niggli, 1928) applied these ideas to crystallographic unit cells, adding the ordering and sign-consistency conditions needed to select a unique canonical representative.

The first practical algorithm implementing Niggli's conditions was given by Křivý & Gruber (1976). Numerical stability issues in that algorithm were subsequently analyzed by Grosse-Kunstleve *et al.* (2004), who proposed modified procedures to improve robustness in floating-point computation. A detailed geometric analysis of the Niggli-reduced region in $\mathbf{G}^6$ and its boundaries is given in Andrews & Bernstein (2014).

3. The $\mathbf{G}^6$ Representation

The metric tensor of a unit cell with basis vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is the 3×3 symmetric positive definite matrix

$$\mathbf{G} = \begin{pmatrix} g_1 & g_6/2 & g_5/2 \\ g_6/2 & g_2 & g_4/2 \\ g_5/2 & g_4/2 & g_3 \end{pmatrix}, \tag{3.1}$$

where

$$\begin{aligned} &g_1 = a^2, \quad g_2 = b^2, \quad g_3 = c^2, \\ &g_4 = 2bc\cos\alpha, \quad g_5 = 2ac\cos\beta, \quad g_6 = 2ab\cos\gamma. \end{aligned} \tag{3.2}$$

The six-component vector

$$\mathbf{g} = (g_1, g_2, g_3, g_4, g_5, g_6) \tag{3.3}$$

defines a point in $\mathbf{G}^6$ (Andrews & Bernstein, 2014). Note that Gruber (1973) introduced the notation $A, B, C, \xi, \eta, \zeta$ for the same quantities, where ξ, η, ζ correspond to our g_4, g_5, g_6; this notation is also used by Krivý & Gruber (1976) and Grosse-Kunstleve *et al.* (2004).

The trace of $\mathbf{G}$ is

$$\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3 = a^2 + b^2 + c^2, \tag{3.4}$$

the sum of squared edge lengths. A unimodular transformation $\mathbf{G}' = \mathbf{M}^T\mathbf{G}\mathbf{M}$, where $\mathbf{M}$ is an integer matrix with $|\det\mathbf{M}| = 1$, changes the basis but not the lattice. Among all unimodular images of a given lattice in $\mathbf{G}^6$, one or more minimize $\mathrm{Tr}(\mathbf{G})$. The secondary conditions of Niggli reduction select a unique cell from among these.

4. Reframing Niggli Reduction as Trace Minimization

4.1. The primary goal

We observe that Niggli reduction can be understood as a two-stage process. The primary stage minimizes $\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3$, finding the most compact possible

cell. The secondary stage applies Niggli's ordering and sign-consistency conditions to select a unique canonical representative from among all cells sharing the minimum trace. This reframing is consistent with — and illuminates — both the original Niggli conditions and the algorithmic formulation of Krivy & Gruber (1976).

Krivy & Gruber (1976) themselves note that in the earlier Buerger algorithm, every step (apart from normalization) strictly decreases $a+b+c$, guaranteeing termination. In the modified Krivy–Gruber algorithm, steps 5–8 are added to enforce the uniqueness conditions, and for these steps $a+b+c$ may remain unchanged. Krivy & Gruber (1976) address this by exhaustive case analysis, asserting that steps 5–8 cover all cases in which $g_1+g_2+g_3$ does not decrease, and that their combined action must eventually lead to a state where it does or where the cell is already reduced. Their termination argument is therefore itself a trace-minimization argument: every execution path either decreases $\mathrm{Tr}(\mathbf{G})$ or provably leads to a state that will.

4.2. Trace degeneracy: multiple cells at the minimum

A critical practical consequence is that more than one distinct cell can achieve the same minimum trace. Figure 1 illustrates this: among 145 unimodular transforms of the input $(g_1, g_2, g_3, g_4, g_5, g_6) = (99.90, 99.98, 99.96, -99.96, 99.87, 0.02)$, three cells share the global minimum $\mathrm{Tr}(\mathbf{G}) = 299.831$, with further clusters of five, five, and three cells at $\mathrm{Tr}(\mathbf{G}) = 299.855$, 299.921, and 299.940 respectively. Niggli's secondary conditions — edge ordering ($g_1 \leq g_2 \leq g_3$), bounds on the off-diagonal components ($|g_4| \leq g_2$, $|g_5| \leq g_1$, $|g_6| \leq g_1$), sign consistency, and tie-breaking rules — are precisely the mechanism by which a unique representative is selected from this degenerate set.

5. Infinite Loops in Niggli Reduction Algorithms

5.1. The loop problem

The possibility of infinite loops in numerical implementations of cell reduction has been recognized since the work of Krivý & Gruber (1976). A simple and effective remedy — a maximum cycle count — was used by one of us as early as 1969 (unpublished). We enumerate here the distinct causes of such loops, since Grosse-Kunstleve *et al.* (2004) observe them empirically but do not analyze their origins.

5.2. Buerger's algorithm and its termination properties

Buerger's algorithm consists of a series of tests, each of which checks whether a specific vector can be shortened, together with the operation that shortens it. If a test fires, the corresponding vector length strictly decreases. In exact arithmetic with a unique minimum, this guarantees termination: since vector lengths are bounded below and each firing produces a strictly shorter vector, no previously visited state can recur.

However, two conditions can break this guarantee even in Buerger's algorithm. First, if two or more cells share the same minimum $\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3$, a step may move from one trace-minimal cell to another of equal trace, and a subsequent step may move back. Since no vector has been shortened, the termination argument does not apply and a cycle is possible. Figure 1 illustrates that multiple trace-minimal cells are common rather than exceptional. Second, floating-point rounding errors can cause a vector that has just been shortened to appear slightly longer than it should, causing the same test to fire again.

5.3. Additional loop causes in the Krivy–Gruber algorithm

Krivý & Gruber (1976) note that in Buerger's algorithm every productive step

shortens a vector. Their modified algorithm adds steps 5–8, whose purpose is to enforce the uniqueness conditions of the Niggli cell rather than to shorten vectors. These steps modify ξ, η, ζ (our g_4, g_5, g_6) without necessarily changing A, B, C (our g_1, g_2, g_3). This introduces a third cause of cycling: a uniqueness step fires and modifies the off-diagonal components, which causes a second step to fire, which restores the conditions for the first step. The cycle can have period 2, 4, 6, or potentially longer. Grosse-Kunstleve *et al.* (2004) observe cycles of period 2, 4, and 6 empirically, for example the repeated sequence A4, A6, A3, A5, A4, A8, but do not analyze why these specific periods arise or whether longer cycles are possible.

A fourth cause applies to severely degenerate cells — those that are nearly planar, nearly collinear, or have extreme length ratios. For such cells the reduction steps may make no numerical progress because the dynamic range of the parameters exceeds the precision of floating-point arithmetic.

In summary the causes are:

1. Multiple cells sharing the same minimum Tr($\mathbf{G}$) (affects Buerger and Krivy–Gruber).

2. Floating-point rounding near a cell-parameter boundary (affects Buerger and Krivy–Gruber).

3. Uniqueness steps that modify g_4, g_5, g_6 without shortening vectors, enabling oscillation between states of equal trace (Krivy–Gruber only).

4. Degenerate cell parameters causing arithmetic underflow or loss of significance (affects all algorithms).

A maximum cycle count guards against all four causes without requiring analysis of which cause is active in a given case.

5.4. The termination approach of Grosse-Kunstleve et al.

Grosse-Kunstleve *et al.* (2004) present two stabilized algorithms. For their modified Krivy–Gruber implementation, termination relies on introducing a tolerance ε into all floating-point comparisons, replacing exact $x < y$ with $x < y - \varepsilon$. This suppresses firings when the cell is within ε of a boundary, reducing the frequency of cause 2 cycles. The value $\varepsilon_{\text{relative}} = 10^{-5}$ is chosen empirically and no proof is given that it bounds accumulated rounding errors in all cases. Causes 1, 3, and 4 are not addressed analytically.

For their separate minimum-reduction algorithm (a simplified Gruber (1973) procedure), they add an explicit stopping criterion: the algorithm terminates when g_1, g_2, g_3 cease to decrease after the N3-false-branch action. This is precisely a trace-minimization criterion — it detects that $\mathrm{Tr}(\mathbf{G})$ has reached its minimum. However, the check is applied only after the N3-false-branch action. Whether cycles that never pass through that branch could evade detection is not addressed.

Grosse-Kunstleve *et al.* (2004) do not separate the contributions of ε and the stopping criterion, as their empirical tests were conducted with both modifications simultaneously active. Whether either alone would suffice is not established.

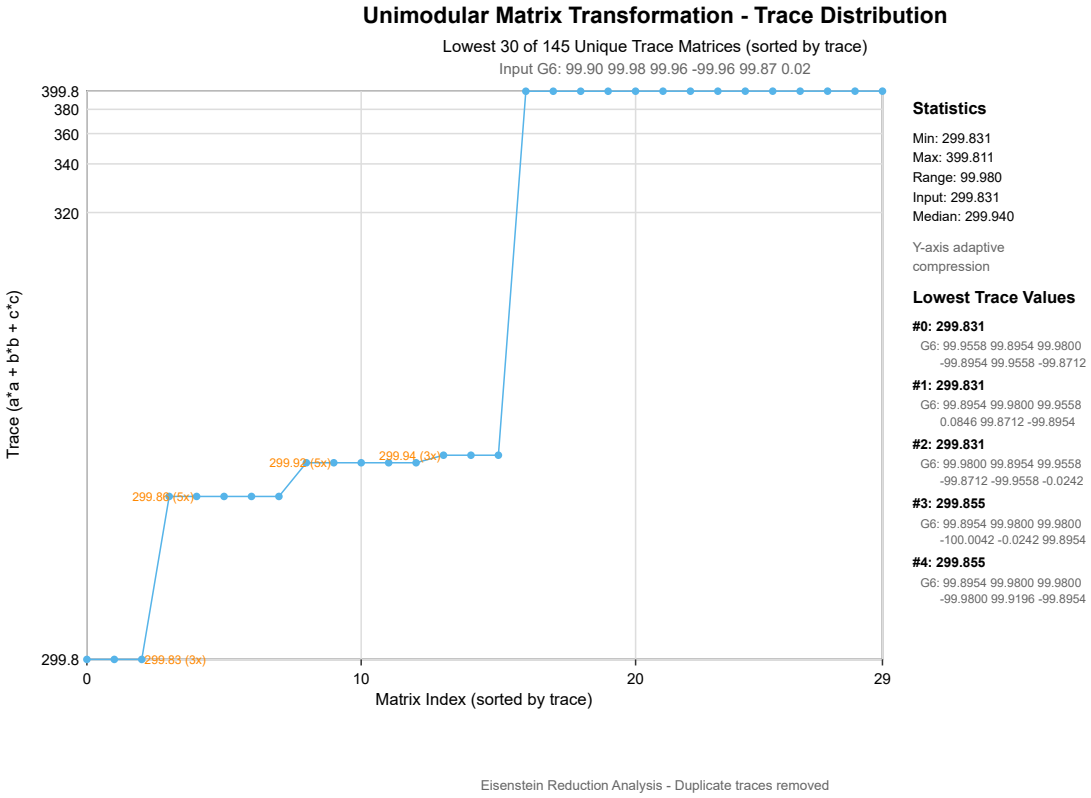


Fig. 1. Trace distribution for the lowest 30 of 145 unimodular transforms of the input cell $G^6 = (99.90, 99.98, 99.96, -99.96, 99.87, 0.02)$. The three leftmost points share the global minimum trace of 299.831; additional clusters at 299.855, 299.921, 299.940 contain five, five, and three cells respectively. Niggli's secondary conditions select a unique reduced cell from among the degenerate trace-minimal candidates.

6. Examples

6.1. Example 1

[To be completed.]

6.2. Example 2

[To be completed.]

7. Summary

Niggli reduction is most naturally understood as a two-stage process: first, minimize $\mathrm{Tr}(\mathbf{G}) = g_1 + g_2 + g_3$ to find the most compact cell; second, apply Niggli's ordering and sign-consistency conditions in $\mathbf{G}^6$ to select a unique canonical representative from among all cells sharing the minimum trace. The secondary conditions are not optional refinements but are essential, because many distinct cells share the same minimum trace value.

The termination argument of Křivý & Gruber (1976) is itself implicitly a trace-minimization argument: every step either decreases $\mathrm{Tr}(\mathbf{G})$ or is shown by case analysis to lead to a state that will. The stopping criterion of Grosse-Kunstleve *et al.* (2004) for their minimum-reduction algorithm makes this explicit. The causes of infinite loops in floating-point implementations are distinct and enumerable: near-boundary floating-point oscillation, interaction between uniqueness steps, near-degeneracy of the trace minimum, and degenerate cell parameters. A maximum cycle count, simple and reliable, guards against all of them.

8. Availability of code

Reduction software and related tools are available at `https://github.com/duck10/LatticeRepLib.git`.

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9. Synopsis

Niggli reduction is reframed as minimization of the trace of the metric tensor, with secondary conditions enforcing uniqueness. Multiple distinct unit cells can share the same minimum trace, and Niggli's conditions select a unique canonical representative from among them.

Appendix
Appendix A. Local unimodular minimality implies the Niggli inequalities

This appendix establishes that a primitive lattice basis which is locally minimal under orientation–preserving unimodular transformations automatically satisfies all Niggli inequalities, up to even permutations and even sign changes of the basis vectors. This provides a geometric explanation for why the Niggli-reduced cell is the canonical representative of a locally minimal basis.

A.1. Notation and assumptions

Let $L \subset \mathbb{R}^3$ be a primitive lattice, and let

$$B = [\,\mathbf{a}\ \mathbf{b}\ \mathbf{c}\,]$$

be a basis matrix whose columns are the basis vectors.

Because L is a primitive crystallographic lattice and chirality must be preserved, we restrict attention to orientation–preserving unimodular integer matrices:

$$U \in \mathrm{SL}_3(\mathbb{Z}), \qquad \det U = +1.$$

Applying U gives another basis

$$B' = B\,U, \qquad (\mathbf{a}', \mathbf{b}', \mathbf{c}') = (\mathbf{a}, \mathbf{b}, \mathbf{c})\,U.$$

Local unimodular minimality. We assume that the basis is *locally minimal* under all elementary orientation–preserving unimodular transformations. Concretely, this means:

- even permutations of $(\mathbf{a}, \mathbf{b}, \mathbf{c})$,
- elementary integer column operations such as $\mathbf{a} \mapsto \mathbf{a} \pm \mathbf{b}$, $\mathbf{b} \mapsto \mathbf{b} \pm \mathbf{c}$, $\mathbf{c} \mapsto \mathbf{c} \pm \mathbf{a}$, provided the resulting matrix lies in $\mathrm{SL}_3(\mathbb{Z})$,
- sign flips of basis vectors only when an even number of vectors are negated (or when combined with an even permutation) so that the resulting matrix remains in $\mathrm{SL}_3(\mathbb{Z})$.

Local minimality means that none of these allowed transformations produces a strictly shorter replacement for the vector it modifies. Equivalently, no allowed unimodular neighbor has a lexicographically smaller metric tuple.

Metric notation. We use the standard G^6 metric representation:

$$g_1 = \mathbf{a}\cdot\mathbf{a}, \qquad g_2 = \mathbf{b}\cdot\mathbf{b}, \qquad g_3 = \mathbf{c}\cdot\mathbf{c},$$

$$g_4 = 2\,\mathbf{b}\cdot\mathbf{c}, \qquad g_5 = 2\,\mathbf{a}\cdot\mathbf{c}, \qquad g_6 = 2\,\mathbf{a}\cdot\mathbf{b}.$$

The factor of 2 is conventional and matches the IUCr definition of the G^6 metric.

Even permutations and even sign flips do not change the diagonal terms and only change the signs of g_4, g_5, g_6. These operations are used only to select a canonical representative and do not affect minimality.

Appendix
Appendix B. Niggli inequalities as consequences of local unimodular minimality

In this appendix we show that the standard Niggli inequalities follow from a precise notion of *local unimodular minimality* under orientation–preserving basis changes. Throughout, we work only with transformations in $\mathrm{SL}_3(\mathbb{Z})$, so that chirality is preserved.

B.1. Local unimodular minimality

Let $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ be a primitive basis of a 3-dimensional lattice, and let

$$(g_1, g_2, g_3, g_4, g_5, g_6) = (a^2, b^2, c^2, 2bc\cos\alpha, 2ac\cos\beta, 2ab\cos\gamma)$$

be the associated metric parameters. The diagonal terms g_1, g_2, g_3 are squared lengths and therefore strictly positive.

Allowed moves. We allow only orientation–preserving basis changes:

$$U \in \mathrm{SL}_3(\mathbb{Z}), \qquad \det U = +1.$$

These include:

- even permutations of $(\mathbf{a}, \mathbf{b}, \mathbf{c})$, whose permutation matrices lie in $\mathrm{SL}_3(\mathbb{Z})$,
- elementary integer column operations such as $\mathbf{a} \mapsto \mathbf{a} \pm \mathbf{b}$, $\mathbf{b} \mapsto \mathbf{b} \pm \mathbf{c}$, etc., provided the resulting matrix lies in $\mathrm{SL}_3(\mathbb{Z})$,
- sign flips of basis vectors only when an *even* number of vectors are negated (or when combined with an even permutation) so that the resulting matrix remains in $\mathrm{SL}_3(\mathbb{Z})$.

Only the off–diagonal terms g_4, g_5, g_6 may change sign under such transformations.

Objective function. We evaluate minimality using the standard Niggli objective

$$\Phi(g_1, g_2, g_3, g_4, g_5, g_6) = (g_1, g_2, g_3, |g_4|, |g_5|, |g_6|)$$

ordered lexicographically. Absolute values appear only on the off–diagonal terms because their signs may change under allowed basis changes, whereas the diagonal terms are squared lengths and remain strictly positive.

Definition. A cell is *locally unimodularly minimal* if for every allowed $U \in \mathrm{SL}_3(\mathbb{Z})$,

$$\Phi(g'_1, \ldots, g'_6) \ \geq_{\mathrm{lex}} \ \Phi(g_1, \ldots, g_6),$$

where $(g'_1, \ldots, g'_6)$ is the metric obtained from the transformed basis.

Canonical sign conventions (e.g. $g_4 \leq 0$) are not consequences of minimality; they are imposed afterwards using transformations that do not decrease Φ.

B.2. Diagonal ordering $g_1 \leq g_2 \leq g_3$

(1) $g_1 \leq g_2$. Assume $g_2 < g_1$. Consider the allowed even permutation

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}) \mapsto (\mathbf{b}, \mathbf{a}, \mathbf{c}),$$

whose matrix has determinant +1. The transformed metric has

$$g_1' = g_2 < g_1,$$

so

$$\Phi(g_1', \ldots, g_6') <_{\mathrm{lex}} \Phi(g_1, \ldots, g_6),$$

contradicting local unimodular minimality. Hence $g_1 \leq g_2$.

(2) $g_2 \leq g_3$**.** The same argument applied to the even permutation $(\mathbf{a}, \mathbf{b}, \mathbf{c}) \mapsto (\mathbf{a}, \mathbf{c}, \mathbf{b})$ shows that $g_2 \leq g_3$ must hold.

Thus local unimodular minimality forces

$$g_1 \leq g_2 \leq g_3.$$

B.3. Off–diagonal bounds

We now show that the inequalities

$$|g_4| \leq g_2, \qquad |g_5| \leq g_1, \qquad |g_6| \leq g_1$$

follow from local unimodular minimality.

(3) $|g_4| \leq g_2$**.** Suppose $|g_4| > g_2$. Consider the allowed transformation

$$\mathbf{b}' = \mathbf{b} - \mathrm{sgn}(g_4)\,\mathbf{c}, \qquad (\mathbf{a}, \mathbf{b}, \mathbf{c}) \mapsto (\mathbf{a}, \mathbf{b}', \mathbf{c}),$$

whose matrix lies in $\mathrm{SL}_3(\mathbb{Z})$. Then

$$\|\mathbf{b}'\|^2 = g_2 + g_3 - 2|g_4|.$$

Since $|g_4| > g_2$ and $g_2 \le g_3$,

$$\|\mathbf{b}'\|^2 < g_3.$$

There are now three possibilities:

- If $\|\mathbf{b}'\|^2 < g_1$, then after reordering by an even permutation the new diagonal triple begins with a strictly smaller first entry.
- If $g_1 \le \|\mathbf{b}'\|^2 < g_2$, then the new diagonal triple has a strictly smaller second entry.
- If $g_2 \le \|\mathbf{b}'\|^2 < g_3$, then the new diagonal triple has a strictly smaller third entry.

In all cases the objective Φ is lexicographically smaller, contradicting minimality. Hence $|g_4| \le g_2$.

(4) $|g_5| \le g_1$**.** If $|g_5| > g_1$, apply the allowed transformation

$$\mathbf{a}' = \mathbf{a} - \mathrm{sgn}(g_5)\,\mathbf{c},$$

whose matrix lies in $\mathrm{SL}_3(\mathbb{Z})$. The same computation shows that $\|\mathbf{a}'\|^2 < g_3$, and the same case analysis yields a lexicographically smaller diagonal triple. Thus $|g_5| \le g_1$.

(5) $|g_6| \le g_1$**.** The argument is identical, using $\mathbf{a}' = \mathbf{a} - \mathrm{sgn}(g_6)\,\mathbf{b}$.

B.4. Sign conventions and tie–breaking

The remaining Niggli conditions (sign conventions and tie–breaking inequalities) do not follow from minimality; they select a canonical representative among metrically equivalent minimal cells.

For example, if $g_4 > 0$ in a region where the convention requires $g_4 \leq 0$, the allowed transformation

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}) \mapsto (\mathbf{a}, -\mathbf{b}, -\mathbf{c}),$$

which has determinant $+1$, changes the sign of g_4 without altering Φ. Thus sign conventions can be imposed without violating minimality.

Similarly, when equalities such as $g_1 = g_2$ occur, Niggli's tie–breaking inequalities select one of several equally minimal cells. These choices do not follow from minimality but are compatible with it.

B.5. Summary

Local unimodular minimality under orientation–preserving basis changes forces:

$$g_1 \leq g_2 \leq g_3, \qquad |g_4| \leq g_2, \qquad |g_5| \leq g_1, \qquad |g_6| \leq g_1.$$

The remaining Niggli conditions (sign conventions and tie–breakers) select a canonical representative among equally minimal cells and are compatible with, but not implied by, minimality.

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